

Preface

Test

Contents

Contents	i
1 Modules Over a Commutative Ring	1
1.1 Categories	1

Chapter 1

Modules Over a Commutative Ring

§ 1.1. Categories

Definition 1.1.1. A *category* $\mathcal{C}$ is a pair of classes $(\text{obj}(\mathcal{C}), \text{Hom}(\mathcal{C}))$, where $\text{obj}(\mathcal{C})$ are the *objects* of $\mathcal{C}$, and $\text{Hom}(\mathcal{C})$ are the *morphisms* of $\mathcal{C}$. Every morphism involves two objects, the *source* (i.e., the domain of the morphism) and the *target* (i.e., the codomain of the morphism). Every object must be identified with a unique *identity morphism* sending the object to itself. The morphisms must satisfy composition, and such a composition must be associative. The set of morphisms between two objects A and B is denoted $\text{Hom}_{\mathcal{C}}(A, B)$.

Example 1.1.1. The category **Set** has as objects sets and morphisms set maps. The category **Group** and **Ring** are defined similarly. The category **Top** has as objects topological spaces and morphisms continuous maps. The category **k-Vec** has as objects k -vector spaces and morphisms linear transformations.

Definition 1.1.2. Let $\mathcal{C}$ be a category.

- (1) An *initial object* of $\mathcal{C}$ is some $I \in \text{obj}(\mathcal{C})$ such that for every $A \in \text{obj}(\mathcal{C})$, there exists a unique morphism $I \rightarrow A$.
- (2) A *terminal object* of $\mathcal{C}$ is some $T \in \text{obj}(\mathcal{C})$ such that for every $B \in \text{obj}(\mathcal{C})$, there exists a unique morphism $B \rightarrow T$.
- (3) A *zero object* of $\mathcal{C}$ is an object that is both initial and terminal.

Example 1.1.2. In **Set**, the emptyset is an initial object and the singletons are terminal objects. There does not exist a zero object in **Set**. In **Ring**, **Group**, **k-Vec**, $\{0\}$ is a zero object.

Definition 1.1.3. Let $\mathcal{C}$ be a category and $\{A_i\}_{i \in I}$ a family of objects in $\mathcal{C}$.

- (1) A *cone* over $\{A_i\}_{i \in I}$ is a pair $(A, \{f_i\}_{i \in I})$ such that $f_i : A \rightarrow A_i$ is a morphism in $\mathcal{C}$ for all i .
- (2) A *cocone* over $\{A_i\}_{i \in I}$ is a pair $(B, \{g_i\}_{i \in I})$ such that $g_i : A_i \rightarrow B$ is a morphism in $\mathcal{C}$ for all i .

Example 1.1.3. Let $\mathcal{C}$ be a category and let $\{A_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. The category **Cone**($\{A_i\}_{i \in I}$) (or just denoted **Cone** if the context is clear) has as objects cones over $\{A_i\}_{i \in I}$ and as morphisms $(A, \{f_i\}_{i \in I}) \rightarrow (A', \{g_i\}_{i \in I})$ making the following diagram commute for each i :

$$\begin{array}{ccc} A & \xrightarrow{\quad} & A' \\ & \searrow f_i & \swarrow g_i \\ & A_i & \end{array}$$

Example 1.1.4. Let $\mathcal{C}$ be a category and let $\{B_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. The category **Cocone**($\{B_i\}_{i \in I}$) (or just denoted **Cocone** if the context is clear) has as objects cocones over $\{B_i\}_{i \in I}$ and as morphisms $(B, \{f_i\}_{i \in I}) \rightarrow (B', \{g_i\}_{i \in I})$ making the following diagram commute for each i :

$$\begin{array}{ccc}
 & B_i & \\
 f_i \swarrow & & \searrow g_i \\
 B & \xrightarrow{\quad} & B'
 \end{array}$$

Definition 1.1.4. Let $\mathcal{C}$ be a category and let $\{A_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. The *product* over $\{A_i\}_{i \in I}$, denoted by $(\prod_{i \in I} A_i, \{\pi_i\}_{i \in I})$, is the terminal object in $\mathbf{Cone}(\{A_i\}_{i \in I})$.

Proposition 1.1.1. Let $\mathcal{C}$ be a category and let $\{A_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. If the product of $\{A_i\}_{i \in I}$ exists, then for any cone $(A, \{\tau_i\}_{i \in I}) \in \text{obj}(\mathbf{Cone})$, there is a unique map $\varphi : A \rightarrow \prod_{i \in I} A_i$ making the following diagram commute for all i :

$$\begin{array}{ccc}
 & \prod_{i \in I} A_i & \\
 \exists! \varphi \nearrow & & \downarrow \pi_i \\
 A & \xrightarrow{\tau_i} & A_i
 \end{array}$$

Definition 1.1.5. Let $\mathcal{C}$ be a category and let $\{B_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. The *coproduct* over $\{B_i\}_{i \in I}$, denoted by $(\coprod_{i \in I} B_i, \{\iota_i\}_{i \in I})$, is the initial object in $\mathbf{Cocone}(\{B_i\}_{i \in I})$.

Proposition 1.1.2. Let $\mathcal{C}$ be a category and let $\{B_i\}_{i \in I}$ be a family of objects in $\mathcal{C}$. If the coproduct of $\{B_i\}_{i \in I}$ exists, then for any cocone $(B, \{\theta_i\}_{i \in I}) \in \text{obj}(\mathbf{Cocone})$, there is a unique map $\varphi : \coprod_{i \in I} B_i \rightarrow B$ making the following diagram commute for all i :

$$\begin{array}{ccc}
 \coprod_{i \in I} B_i & & \\
 \uparrow \iota_i & \searrow \exists! \varphi & \\
 B_i & \xrightarrow{\theta_i} & B
 \end{array}$$

Example 1.1.5. Consider the collection $\{\mathbb{Z}_n\}_{n \in \mathbb{N}} \subseteq \text{obj}(\mathbf{Group})$. Let $\pi_n : \mathbb{Z} \rightarrow \mathbb{Z}_n$ be the natural quotient maps. There does not exist a map of cones $(\mathbb{Z}, \{\pi_n\}_{n \in \mathbb{N}}) \rightarrow 0$.